TOOLKIT

Master Glossary

Every new word from every day, A to Z.

Every new word from every day, A to Z. The tag (e.g. `W1D1`) shows the day it was introduced. This file is rebuilt automatically from the daily glossaries.

0 Lag 0

- **Plain English:** The first bar of every correlogram. Always exactly 1, because a series matches itself perfectly.
- **Naija analogy:** Asking "do you look like yourself?" Of course you do. Ignore that question.
- **Example:** The grey bar at the far left of every correlogram today.

W2D2

1 Lag-1 autocorrelation

- **Plain English:** How much today matches yesterday.
- **Naija analogy:** If it rained heavily today, there's a good chance it rains tomorrow too in July.
- **Example:** `s.autocorr(lag=1)` gives 0.58 for throughput.

W2D2

3 Three-cycle rule

- **Plain English:** Have at least three full cycles of data before you trust a seasonal pattern.
- **Naija analogy:** One friend says the new buka is sweet: gist. Three different friends: pattern.
- **Example:** 36 monthly rows = exactly 3 rainy seasons: the minimum.

W2D1

🗣 4-sentence finding

- **Plain English:** Situation → Observation → Implication → Recommendation. A whole report in four sentences.
- **Naija analogy:** How a good mechanic talks: “Your car is old (S). The brake pads are finished (O). You could crash (I). Change them today (R).”
- **Example:** The last slide of today's deck. The recommendation always comes last.

W1D2

➕ Additive model

- **Plain English:** observed = trend + seasonal + residual. The seasonal swing stays the same size even when the level rises.
- **Naija analogy:** December transport fare goes up by a fixed ₦100 every year, whatever the normal fare is.
- **Example:** `seasonal_decompose(series, model="additive", period=12)`

W2D1

🤖 AI button (Gemini)

- **Plain English:** Colab's built-in AI helper. Great for explaining code and errors.
- **Naija analogy:** A very smart friend who talks a lot. Never tell him company secrets.
- **Example:** OK to ask: "Explain this error." Never paste real NIGCOMSAT telemetry.

W1D1

🚨 Alert line (threshold)

- **Plain English:** A value that, once crossed, means someone should act. Real ones are set by engineers.
- **Naija analogy:** The fuel light on your car's dashboard.
- **Example:** The lab's 97 Ah line is **made up for practice**.

W2D3

🎚️ Alpha

- **Plain English:** The setting for exponential smoothing. High alpha = follows the data closely and reacts fast. Low alpha = smooth and slow.
- **Naija analogy:** High alpha believes every WhatsApp broadcast immediately. Low alpha waits for three people to confirm.
- **Example:** alpha 0.5 hugs the daily line; alpha 0.05 is very calm.

W1D3

📏 .asfreq()

- **Plain English:** Declares the rhythm of your data (e.g. daily) and creates an empty slot for any missing period.
- **Naija analogy:** A class register with every school day printed on it. An absent student shows as a blank, not a missing page.
- **Example:** `ka.asfreq("D")`. Missing days appear as `NaN`.

W1D1

🧠 Autocorrelation

- **Plain English:** How much a series looks like its own shifted copy. One number from −1 to +1.
- **Naija analogy:** How much your Monday looks like last Monday. Same traffic, same stress, same suya spot.
- **Example:** Throughput: lag 1 = 0.58, lag 7 = 0.99.

W2D2

✅ Availability

- **Plain English:** The percentage of time a link worked properly.
- **Naija analogy:** Out of 24 hours, how many did your network actually give you full bars?
- **Example:** 99.6% availability = about 6 minutes of bad service in a day.

W1D1

🔪 Baseline

- **Plain English:** The simplest reasonable method, the one every fancier method must beat on the same test.
- **Naija analogy:** Grandma's weather forecast: “tomorrow will be like today”. Beat her or go home. 👵🏾
- **Example:** “Same weekday last week” as a forecast for throughput.

W2D3

👵🏾 Baseline forecast

- **Plain English:** The simplest possible forecast. Any clever model must beat it, or it's not worth using.
- **Naija analogy:** Grandma's weather forecast: "tomorrow will be like today." Hard to beat!
- **Example:** "Same as yesterday" and "same weekday last week".

W2D2

🎯 Brief

- **Plain English:** One of the five project types you can choose (A–E), each with its own operational question.
- **Naija analogy:** The topic the lecturer approves for your final-year project.
- **Example:** Brief B: “When will the battery be too tired to carry us through eclipse?”

W2D3

🥁 Cadence (frequency)

- **Plain English:** How often a reading is recorded: hourly, daily, weekly, monthly.
- **Naija analogy:** How often your landlord comes for rent: monthly? yearly? (Hopefully not weekly 😅)
- **Example:** `bus_power_telemetry` is hourly; `ka_band_availability` is daily.

W1D1

🔋 Capacity (amp-hours, Ah)

- **Plain English:** How much charge a battery can still hold. It shrinks as the battery ages.
- **Naija analogy:** Your old phone that used to last all day and now dies before lunch.
- **Example:** `est_capacity_ah` falls from 103.3 Ah to 96.6 Ah over three years.

W2D3

🧱 Cell (code cell / text cell)

- **Plain English:** One block in a notebook. A **code cell** runs Python; a **text cell** is just for reading.
- **Naija analogy:** One stop on a danfo route. You can't jump to Oshodi without passing the stops before it.
- **Example:** `print("Hello")` lives in a code cell. These instructions live in text cells.

W1D1

🔮 Centred window

- **Plain English:** A window with today in the middle: half the days are in the past, half in the **future**.
- **Naija analogy:** Marking a WAEC script after you've seen the answer sheet. Looks perfect, but it's cheating if you use it live.
- **Example:** `s.rolling(30, center=True).mean()` is empty for the last 15 days.

W1D3

🚩 Checkpoint

- **Plain English:** A short progress check at a fixed point in the course.
- **Naija analogy:** Police checkpoints on the expressway, but these ones help you. 😄
- **Example:** Week 3 · Day 3: bring your unit head's reply and your profiled data.

W2D3

🧩 Component

- **Plain English:** One of the parts a decomposition produces.
- **Naija analogy:** One instrument in a highlife band. The song is all of them playing together.
- **Example:** `parts.trend`, `parts.seasonal`, `parts.resid`

W2D1

🟦 Confidence band

- **Plain English:** The shaded area in a correlogram. Bars inside it could easily be chance.
- **Naija analogy:** The "e no too matter" zone: small differences you shouldn't argue about.
- **Example:** For 730 days of data the band is about ±0.07.

W2D2

🔗 Correlation

- **Plain English:** A number from −1 to +1 showing how much two series move together. It does **not** tell you which one causes the other.
- **Naija analogy:** Umbrella sales and traffic jams both rise when it rains. Umbrellas don't cause traffic.
- **Example:** Rain and uplink power: correlation 0.97.

W1D2

🧭 Correlation ≠ cause

- **Plain English:** Moving together doesn't prove one causes the other. Even when there IS a cause, check which direction it runs.
- **Naija analogy:** Umbrella sales and rain go together. Selling umbrellas doesn't bring rain.
- **Example:** Rain and uplink power: r = 0.97, because rain makes the station turn up its power, not the other way round.

W2D2

🔗 Correlation (r)

- **Plain English:** How closely two different series move together, from −1 to +1.
- **Naija analogy:** How often Mama Ngozi and Mama Bisi show up at the same parties.
- **Example:** `a.corr(b)`

W2D2

Correlogram (ACF plot)

- **Plain English:** A bar chart with one bar per lag, showing the autocorrelation at each.
- **Naija analogy:** A report card for memory: one score for yesterday, one for last week, one for last month.
- **Example:** `plot_acf(s.dropna(), lags=35)`.

W2D2

CSV file

- **Plain English:** A plain-text table where commas separate the columns.
- **Naija analogy:** A list written in a jotter with commas instead of lines.
- **Example:** `date,tickets_raised` / `2024-01-01,8`

W1D1

Cycle

- **Plain English:** Ups and downs that repeat, but **with no fixed length**.
- **Naija analogy:** Business boom years and slow years. Nobody can tell you the exact date of the next boom.
- **Example:** Demand strong for 2 years, weak for 3, then rising again.

W1D1

Daily average (mean)

- **Plain English:** Add up all the readings in a day and divide by how many. Shows the typical level.
- **Naija analogy:** "On average, NEPA gave us 20 hours today", which hides that the 4 missing hours were during the match.
- **Example:** `rf["link_margin_db"].resample("D").mean()` : never below 4.9 dB in August.

W1D2

Daily minimum

- **Plain English:** The single lowest reading of each day: the worst moment.
- **Naija analogy:** The one hour you sat on Third Mainland Bridge: that's the hour you remember.
- **Example:** `resample("D").min()` showed an outage on 11 of 31 days.

W1D2

📋 DataFrame

- **Plain English:** pandas' name for a table: rows and named columns.
- **Naija analogy:** The class register: one row per student, one column per day.
- **Example:** `ka` is a DataFrame with 1,096 rows.

W1D1

📉 dB (decibel)

- **Plain English:** A unit for signal strength on a "times" scale: +3 dB is about double, −3 dB about half.
- **Naija analogy:** Like Richter scale for earthquakes: small number changes mean big real changes.
- **Example:** The worst hour in August hit −4.3 dB of margin.

W1D2

🔧 Decomposition

- **Plain English:** Splitting one time series into separate parts: trend, seasonal and residual.
- **Naija analogy:** Tasting the jollof and naming each ingredient: rice, tomato base, and that one pepper that surprised you.
- **Example:** `seasonal_decompose(kam["availability_pct"], period=12)`

W2D1

🪫 Depth of discharge

- **Plain English:** How far the battery was drained, as a percentage of what it holds.
- **Naija analogy:** How low your inverter battery went during last night's blackout.
- **Example:** Under 1% on normal days, up to about 22% on the deepest eclipse days.

W2D3

➖ Differencing (.diff())

- **Plain English:** Replacing each value with the change since the previous step. It removes the trend.
- **Naija analogy:** Instead of asking "how much money do you have?", asking "how much did you gain or lose this week?"
- **Example:** Suya vs generators: 0.97 on levels, 0.06 on changes.

W2D2

☝🏾 Dual axis (avoid it)

- **Plain English:** Two different y-axes on one chart, one on each side. It lets you make any two lines look related.
- **Naija analogy:** Two referees on one pitch with different rule books.
- **Example:** Rain and uplink power go in two stacked charts instead.

W1D2

👯 Duplicate timestamp

- **Plain English:** The same time appearing twice: usually an export mistake.
- **Naija analogy:** Being charged twice on one POS transaction.
- **Example:** Two rows both saying `2024-03-06`.

W1D1

🌒 Eclipse season

- **Plain English:** The weeks around each equinox (March and September) when the Earth's shadow crosses the satellite's orbit for up to 72 minutes a day.
- **Naija analogy:** NEPA's scheduled load-shedding season, but for the satellite. It switches to battery.
- **Example:** Eclipse hours sit at 21:00 UTC from 1 September to 16 October in the bus-power file.

W2D1 W2D3

🔋 End-of-discharge voltage

- **Plain English:** The battery's voltage at the end of an eclipse, when it has worked hardest. Lower means a deeper discharge or an older battery.
- **Naija analogy:** How much power is left in your inverter battery just before NEPA brings light back.
- **Example:** `eod_voltage_v` dips in March and September and slowly drifts down over the years.

W2D1 W2D3

🌍 Equinox

- **Plain English:** The two days a year (around 21 March and 23 September) when day and night are equal. Eclipse seasons are centred on them.
- **Naija analogy:** The two fixed “owambe Saturdays” of the satellite calendar: you can plan for them years ahead.
- **Example:** The seasonal bumps in the lab land in March and September.

W2D3

Exponential smoothing (`.ewm`)

- **Plain English:** A smoother that gives the newest values the most weight and older values less and less.
- **Naija analogy:** How you rate a footballer: last week's match counts most, matches from last season count a little.
- **Example:** `s.ewm(alpha=0.3).mean()`

W1D3

Finding (4-sentence)

- **Plain English:** A short write-up in four sentences: Situation, Observation, Implication, Recommendation.
- **Naija analogy:** How a good doctor explains: what's happening, what the test shows, what it means, what to do.
- **Example:** "Weekends average 3.4 tickets a day vs 7.1 on weekdays" is an Observation.

W2D3

Finding title

- **Plain English:** A chart title that states what the chart shows, ideally with a number.
- **Naija analogy:** A newspaper headline: "Eagles win 2–0", not "Football match".
- **Example:** "Ka-band link went down on 11 of 31 August days".

W1D2

Gap

- **Plain English:** A period with no reading at all, e.g. a missing day.
- **Naija analogy:** The days in your diary when you were "too busy" to write.
- **Example:** In B5 we found 3 missing days in March 2024.

W1D1

GEO (geostationary) satellite

- **Plain English:** A satellite about 36,000 km up that moves with the Earth's spin, so it stays over the same spot.
- **Naija analogy:** A security light on a high pole: it always shines on the same compound.
- **Example:** NigComSat-1R sits at 42.5° East.

W1D1

🗺️ Geopolitical zones

- **Plain English:** Nigeria's six regional groupings: North West, North East, North Central, South West, South East, South South.
- **Naija analogy:** The six "houses" at inter-house sports.
- **Example:** The `region` column in the VSAT file.

W1D2

💻 Google Colab

- **Plain English:** A free website where you write and run Python. Google's computer does the work; yours just shows the screen.
- **Naija analogy:** A cyber café computer that lives in your browser, never runs out of fuel and never asks for ₦200 per hour.
- **Example:** You open `W1D1-exercise.ipynb` in Colab and run it without installing anything.

W1D1

📌 Index (DatetimeIndex)

- **Plain English:** The row labels of a table. When they're dates, pandas can do time tricks with them.
- **Naija analogy:** The page numbers of a book. Put dates there and you can jump straight to "August 2024."
- **Example:** `index_col="timestamp"` makes the dates the index.

W1D1

💯 Index to a baseline

- **Plain English:** Divide every value by a starting value and multiply by 100, so everything starts at 100 and you see *change*.
- **Naija analogy:** Comparing students by how much they improved since last term, not by raw score.
- **Example:** `wide / wide.iloc[0] * 100` : South East fell furthest from its own start.

W1D2

🔎 Investigation

- **Plain English:** Answering one question with data, from loading the file to writing the conclusion.
- **Naija analogy:** A mechanic's full diagnosis, not just "e don spoil".
- **Example:** "Is our battery getting tired?": load → profile → plot → smooth → decompose → ACF → finding.

W2D3

📶 Ka-band

- **Plain English:** A high-frequency radio band that carries lots of data but is very sensitive to rain.
- **Naija analogy:** A sports car: very fast, but it hates bad roads (here: heavy rain).
- **Example:** The Ka-band Nigeria spot beam in today's dataset.

W1D1

🪞 Lag

- **Plain English:** How many steps back you look. Lag 1 = one step ago, lag 7 = seven steps ago.
- **Naija analogy:** "Last week this time" when people gist about what happened last week at this same hour.
- **Example:** On daily data, lag 7 = the same weekday last week. On hourly data, lag 24 = the same hour yesterday.

W2D2

⏰ Lag (delay)

- **Plain English:** How late a smoothed line reacts to a real change. For a trailing window, about half the window.
- **Naija analogy:** News reaching the village two weeks after it happened in Abuja.
- **Example:** The trailing 30-day peak came 14 days after the centred one.

W1D3

⛽ Link margin (dB)

- **Plain English:** How much signal is left above the minimum the receiver needs. Below 0 dB the link is down.
- **Naija analogy:** Fuel above the reserve line. Positive: you can still drive. Zero: you're pushing the car. 🚗
- **Example:** `link_margin_db` in `rf_link_quality_hourly`.

W1D2

🪄 Logarithm (log)

- **Plain English:** A maths function that turns multiplying into adding. `np.log()` in Python.
- **Naija analogy:** A translator who turns "times" talk into "plus" talk so everybody understands.
- **Example:** `np.log(demo["multiplicative"])`: the swing stops growing.

W2D1

🔪 Long format

- **Plain English:** One row per group per time: many rows, few columns. How databases usually export.
- **Naija analogy:** A class register written as one line per student per day.
- **Example:** `vsat_uptime_by_region_weekly`: 312 rows = 52 weeks × 6 zones.

W1D2

🔪 MAE (mean absolute error)

- **Plain English:** The average size of a forecast's mistakes, ignoring whether they were too high or too low.
- **Naija analogy:** How many minutes late your friend usually is, on average. 😅
- **Example:** `(s - s.shift(7)).abs().mean()`

W2D2

〰 Moving average (centred)

- **Plain English:** The average of a window of values around each point. Classic decomposition uses one to find the trend, so the trend is missing for half a window at each end.
- **Naija analogy:** Judging a student by the average of the tests around each week. You can't do it for the first and last weeks.
- **Example:** The Ka-band trend is empty for the first and last 6 months.

W2D1

🧱 Moving (rolling) average

- **Plain English:** The average of the last N values, recalculated every day as the window moves forward.
- **Naija analogy:** Your average fuel spend over the last 7 days. One big fill-up on Monday doesn't make you panic.
- **Example:** The 7-day average of NOC tickets sits around 6 a day in the dry season.

W1D3

✖ Multiplicative model

- **Plain English:** observed = trend × seasonal × residual. The seasonal swing grows as the level grows.
- **Naija analogy:** December fare goes up 20%. When fares are higher, 20% is more naira.
- **Example:** The `multiplicative` column in `additive_multiplicative_demo`: swing grows from about 0.29 to 0.54.

W2D1

🏭 Named process

- **Plain English:** A specific, real task your unit does regularly, with an owner. Not a general topic.
- **Naija analogy:** Not "food": "Mama's Saturday jollof for the street party".
- **Example:** "The weekly battery-health review on the bus subsystem."

W2D3

🧠 NameError

- **Plain English:** Python's way of saying "I don't know this name." Usually you skipped a cell, restarted, or forgot quotes.
- **Naija analogy:** Calling someone "Chief" at a party when nobody has introduced him to you yet.
- **Example:** `NameError: name 'ka' is not defined` → run the cells above.

W1D1

➖ NaN

- **Plain English:** "Not a Number". pandas' symbol for an empty or missing value.
- **Naija analogy:** The empty seat at a wedding where the guest didn't come.
- **Example:** After `.asfreq("D")`, the 3 deleted days show `NaN`.

W1D1

🎧 NOC (Network Operations Centre)

- **Plain English:** The team that watches the network 24/7 and opens a ticket for every fault.
- **Naija analogy:** The control room of a big estate, where security watches all the CCTV.
- **Example:** `noc_tickets_daily.csv` counts their tickets.

W1D3

🎲 Noise

- **Plain English:** Random wobble that means nothing. Don't react to it.
- **Naija analogy:** Weighing yourself after eating amala: 2 kg up doesn't mean you're getting fat. 😄
- **Example:** NOC tickets 38 one week, 41 the next = the same week twice.

W1D1

☕ Noise (again)

- **Plain English:** Random week-to-week wobble with no meaning. Don't react to it.
- **Naija analogy:** Your weight after a heavy plate of pounded yam. Wait a week before you worry.
- **Example:** 38 vs 41 tickets a week, when the mean is about 36 and the usual swing is about 5.

W1D3

📓 Notebook

- **Plain English:** A document that mixes reading (text) and code you can run, in one page.
- **Naija analogy:** A recipe book where you can cook each step right on the page and taste it immediately.
- **Example:** Files ending in `.ipynb` are notebooks.

W1D1

🐼 pandas

- **Plain English:** The Python library for working with tables of data.
- **Naija analogy:** Microsoft Excel, but you control it by typing instructions, so it never gets tired.
- **Example:** `import pandas as pd`, then `pd.read_csv(...)`.

W1D1

🗒️ parse_dates

- **Plain English:** Tells pandas "this column is a date, not just text."
- **Naija analogy:** Telling the gateman, "this is not just any paper, it's my ID card."
- **Example:** `pd.read_csv(..., parse_dates=["timestamp"])`

W1D1

✂️ Partial period

- **Plain English:** A bucket at the edge of the data that doesn't have all its days.
- **Naija analogy:** Judging a football match at half-time and saying the team scored few goals.
- **Example:** The last weekly bucket has only 2 days, so it shows just 9 tickets.

W1D3

🥁 Period

- **Plain English:** How many rows until the seasonal pattern repeats. It comes from the process, not from a default.
- **Naija analogy:** How often your esusu (contribution) group meets: weekly means 7 days, monthly means 12 times a year.
- **Example:** 24 for hourly bus power, 7 for daily tickets, 12 for monthly availability.

W2D1 W2D2

🧩 Persistence (memory)

- **Plain English:** A series "has memory" when its past values help you guess its next values.
- **Naija analogy:** A neighbour who has been borrowing sugar every week will likely knock again next week. 😅
- **Example:** Throughput has strong weekly memory; random noise has none.

W2D2

🔄 pivot

- **Plain English:** The pandas command that reshapes long → wide.
- **Naija analogy:** Rearranging wedding guests from one long queue into tables by family.
- **Example:** `vsat.pivot(index="week_start", columns="region", values="uptime_pct")`

W1D2

📐 plot_acf / statsmodels

- **Plain English:** `statsmodels` is a Python library for statistics; `plot_acf` is its correlogram function. Colab already has it.
- **Naija analogy:** A specialist mechanic: pandas is the general workshop, statsmodels is the one who knows engines.
- **Example:** `from statsmodels.graphics.tsaplots import plot_acf`

W2D2

🚨 Priority-1 ticket

- **Plain English:** The most urgent kind of fault ticket. Rare, but serious.
- **Naija analogy:** A call from your MD at 11pm: it doesn't happen often, but when it does, everybody moves.
- **Example:** About 1 a day, and 17% of days have none.

W1D3

🧑🏾‍💼 Process owner

- **Plain English:** The person responsible for that process, who will use or approve your tool.
- **Naija analogy:** The oga at the top of that particular desk.
- **Example:** The shift lead who runs NOC fault escalation.

W2D3

🔍 Profiling

- **Plain English:** Quick checks before you trust a dataset: rows, date range, gaps, duplicates, missing values.
- **Naija analogy:** Inspecting a tokunbo car before you pay: engine, tyres, papers.
- **Example:** `len(ka)`, `ka.index.min()`, `ka.isna().sum()`.

W1D1

🌧️ Rain fade

- **Plain English:** Heavy rain absorbs the satellite signal, so quality drops or the link goes down.
- **Naija analogy:** DStv showing "No signal" during a heavy Lagos downpour.
- **Example:** The yearly dips in availability, worst in July and August.

W1D1 W1D2

🚫 Random-split leakage

- **Plain English:** If you split time data randomly, the test set's near-copies sit in the training set, so the answers leak.
- **Naija analogy:** WAEC expo: if you saw the questions before the exam, your score is fake.
- **Example:** Day 8 in test, days 1 and 15 (same weekday, r = 0.99) in training.

W2D2

📦 Raw data

- **Plain English:** The original export from a system, before you clean or summarise it.
- **Naija analogy:** Unpeeled yam straight from the market.
- **Example:** Never commit raw NIGCOMSAT exports to GitHub: commit only your code.

W2D3

🗃️ Repository (repo)

- **Plain English:** A project folder tracked by Git, often stored on GitHub, holding your code and its history.
- **Naija analogy:** Your project file at the registry, with every version stamped and dated.
- **Example:** Your final project's code, `requirements.txt` and `README.md` live in a repo.

W2D3

🔁 resample

- **Plain English:** Change the time step of a series (hourly → daily) and choose how to summarise each new step.
- **Naija analogy:** Turning a daily sales diary into a weekly report: you must decide to add, average or take the best day.
- **Example:** `.resample("D").min()`

W1D2 W1D3

🎲 Residual

- **Plain English:** What's left after the trend and season are removed. Mostly noise, but big residuals are surprises worth investigating.
- **Naija analogy:** After you've explained the traffic by "it's Monday" and "it's raining", whatever delay is left is the keke that broke down.
- **Example:** In August, the biggest residual points straight at an injected solar-array dropout.

W2D1

↩️ Response (vs cause)

- **Plain English:** Something that changes *because of* something else.
- **Naija analogy:** You carry an umbrella because it's raining. The umbrella didn't bring the rain.
- **Example:** Uplink power is a response to rain.

W1D2

🔒 Restricted data

- **Plain English:** Real NIGCOMSAT data that must not leave NIGCOMSAT systems or go into outside tools without permission.
- **Naija analogy:** Your ATM PIN: useful, but you don't paste it into WhatsApp.
- **Example:** Real telemetry, RF records and network logs. Course data is synthetic and safe.

W2D3

🛡️ Robust

- **Plain English:** A method that doesn't let a few extreme values pull the result around.
- **Naija analogy:** One bad experience at one buka shouldn't make you blacklist every buka in Lagos.
- **Example:** With one fake disaster August, robust STL keeps August's seasonal effect near −1.2 instead of −3.0.

W2D1

📉 Rolling standard deviation

- **Plain English:** How much the values jumped around inside the window. A measure of wobble.
- **Naija analogy:** How much your neighbour's generator noise goes up and down in one evening.
- **Example:** `s.rolling(7).std()`

W1D3

🌊 Rolling window

- **Plain English:** A fixed number of rows that slides along the series one row at a time. You calculate something (usually the average) inside it.
- **Naija analogy:** A danfo with exactly 7 seats: every time one passenger enters at the front, the oldest one drops at the back.
- **Example:** `tk["tickets_raised"].rolling(7).mean()`

W1D3

🔄 Runtime / session

- **Plain English:** The live connection to Google's computer, plus everything it currently remembers.
- **Naija analogy:** A phone call. Hang up (restart) and the other person forgets what you were saying.
- **Example:** *Runtime → Restart session* wipes the memory. Run STEP 0 again after it.

W1D1

💾 Save a copy in Drive

- **Plain English:** Makes your own editable copy of the notebook in your Google Drive.
- **Naija analogy:** Photocopying the class notes before writing on them, so you don't scribble on the original.
- **Example:** *File → Save a copy in Drive*. Your copy lands in Drive → *Colab Notebooks*.

W1D1

🔁 Seasonal component

- **Plain English:** The pattern that repeats exactly every period: the same 12 numbers every year, or the same 7 numbers every week.
- **Naija analogy:** Rush hour at 7am every weekday, like clockwork.
- **Example:** August is about 1.1 points below the yearly average every year.

W2D1

🔧 seasonal_decompose

- **Plain English:** The classic decomposition function. Simple and fast, but it assumes the season looks the same every cycle.
- **Naija analogy:** A tailor who uses one measurement for you every year, even if you've added weight. 😅
- **Example:** `parts = seasonal_decompose(series, model="additive", period=7)`

W2D1

🗓️ Seasonal naive ("same weekday last week")

- **Plain English:** Forecast each day with the value from one period ago.
- **Naija analogy:** Expecting this Friday's traffic to be like last Friday's traffic.
- **Example:** For throughput its average error is 0.062 Gbps, about 5× better than "same as yesterday" (0.338).

W2D2

🌧️ Seasonality

- **Plain English:** A pattern that repeats on a **fixed calendar**: every day, every week, every year.
- **Naija analogy:** Harmattan every December, rainy season every year, traffic every weekday morning.
- **Example:** Ka-band availability dips every August.

W1D1

📈 Shared trend

- **Plain English:** Two series both rising (or falling) over time for their own reasons, which makes them look linked.
- **Naija analogy:** Your salary and the price of garri both rose over 5 years. One didn't cause the other.
- **Example:** Amplifier temperature and RF drive both drift up over two years.

W2D2

⏪ .shift()

- **Plain English:** The pandas command that makes a lagged copy by sliding every value down.
- **Naija analogy:** Everyone in the queue moving one seat back so a latecomer can sit in front.
- **Example:** `s.shift(1)` puts yesterday's value on today's row.

W2D2

⌨️ Shift + Enter

- **Plain English:** The keyboard shortcut that runs the current cell and moves to the next.
- **Naija analogy:** "Next please!" at the bank queue.
- **Example:** Click a cell, press Shift + Enter, see the output.

W1D1

📜 SLA (Service Level Agreement)

- **Plain English:** The promise to a customer, e.g. "the link will be up 99.5% of the time".
- **Naija analogy:** The landlord's promise of "24 hours light and water" in the house advert. 😅
- **Example:** "Did we breach the SLA?" is answered with daily **minimums**, not averages.

W1D2 W1D3

🦕 Slow decay

- **Plain English:** Bars that start high and fade gently, lag after lag. It's the fingerprint of a trend.
- **Naija analogy:** Lagos rent: this year looks like last year, which looked like the year before… all going one way.
- **Example:** Transponder utilisation: 0.90, 0.81, 0.73, 0.64…

W2D2

🖼️ Small multiples

- **Plain English:** A grid of small, identical charts, one per group, sharing the same y-axis.
- **Naija analogy:** Passport photos of the whole family: same size, same background, so you can compare faces.
- **Example:** Six mini charts, one per geopolitical zone, `sharey=True`.

W1D2

🌀 Smoothing

- **Plain English:** Removing the small, fast wiggles so the slower, real movement is easier to see.
- **Naija analogy:** Ironing a shirt: the shape is still there, the creases are gone.
- **Example:** The 7-day line shows the rainy-season rise hidden inside the daily zig-zag.

W1D3

🌞 Solar-array current

- **Plain English:** The electric current coming from the satellite's solar panels. About 27.8 A in full sunlight.
- **Naija analogy:** The output of your rooftop solar panel on a sunny afternoon.
- **Example:** A sudden drop to about 16 A with no eclipse is a fault: a "string dropout".

W2D1

🍝 Spaghetti chart

- **Plain English:** Too many lines on one chart. Nobody can follow any of them.
- **Naija analogy:** Ten danfo routes drawn on one map with the same blue pen.
- **Example:** Six zones on one axis: try finding South South.

W1D2

🔢 Sparse (small) counts

- **Plain English:** Counts that are usually 0, 1 or 2. One extra event is a big jump, so they need wider windows.
- **Naija analogy:** Counting how many times NEPA brings light at 3am. Mostly zero, sometimes one.
- **Example:** Priority-1 tickets per day.

W1D3

📈 Spike

- **Plain English:** A bar much taller than its neighbours, sticking above the shaded band.
- **Naija analogy:** The one day in the month when everyone rushes to the bank: salary day.
- **Example:** Throughput spikes at lags 7, 14, 21, 28, 35.

W2D2

🔦 Spot beam

- **Plain English:** A satellite beam aimed at one region, not the whole continent.
- **Naija analogy:** A torchlight pointed at one spot instead of a bulb lighting the whole room.
- **Example:** “Nigeria spot beam” in the `beam` column.

W1D1

🎭 Spurious correlation

- **Plain English:** A strong correlation between two things that are NOT really linked.
- **Naija analogy:** Suya sales and generator sales both rise toward December. Suya doesn't sell generators. 😄
- **Example:** Made-up suya vs generators: r = 0.97.

W2D2

📦 statsmodels

- **Plain English:** A Python library for statistics and time-series models. Already installed in Colab.
- **Naija analogy:** The big toolbox in the mechanic's shop. pandas is the spanner; statsmodels has the special tools.
- **Example:** `from statsmodels.tsa.seasonal import seasonal_decompose`

W2D1

🧠 STL

- **Plain English:** A smarter decomposition (Seasonal-Trend decomposition using LOESS) that lets the season change slowly over time.
- **Naija analogy:** A tailor who measures you again before every new outfit.
- **Example:** `STL(series, period=24, robust=True).fit()`

W2D1

🧪 Synthetic data

- **Plain English:** Made-up data designed to behave like the real thing. It's safe to share and safe to use with AI.
- **Naija analogy:** A mock exam: same style as WAEC, no real stakes.
- **Example:** All course datasets are synthetic, built to behave like NigComSat-1R.

W1D1

🚧 Threshold line

- **Plain English:** A line drawn at the level that matters (0 dB, the SLA target), so readers instantly see breaches.
- **Naija analogy:** The red line on a fuel gauge.
- **Example:** `plt.axhline(0, color="red", ls="--")`

W1D2

📶 Throughput (Gbps)

- **Plain English:** How much data a link carries, in gigabits per second.
- **Naija analogy:** How many cars pass through the toll gate per minute.
- **Example:** Over two years, about 3.9 Gbps on an average weekday and 3.0 at weekends.

W2D2

⏱️ Time series

- **Plain English:** Measurements recorded one after another in time, where the **order matters**.
- **Naija analogy:** Your NEPA light diary: "06:00 light came, 09:00 light gone…" Shuffle it and it means nothing.
- **Example:** Daily Ka-band availability from 2023 to 2025.

W1D1

🕐 Timestamp

- **Plain English:** The date and time attached to a reading.
- **Naija analogy:** The time printed on your POS receipt.
- **Example:** `2025-08-01T21:00` = 1 August 2025, 9pm UTC.

W1D1

⬅️ Trailing window

- **Plain English:** A window that ends today and only looks at the past. pandas does this by default.
- **Naija analogy:** Driving using only your rear-view mirror: safe information, but always a bit late.
- **Example:** `s.rolling(30).mean()` is trailing.

W1D3

📡 Transponder

- **Plain English:** A channel on the satellite that receives a signal and sends it back down. NigComSat-1R has 28 active ones.
- **Naija analogy:** One lane on a busy expressway. More lanes = more traffic carried.
- **Example:** `transponder_utilisation_monthly.csv` tracks how busy they are.

W1D1

📈 Trend

- **Plain English:** The long-term direction: up, down or flat.
- **Naija analogy:** Lagos population: more people every year.
- **Example:** Transponder utilisation climbing from about 50% to about 78%.

W1D1

📈 Trend component

- **Plain English:** The slow, long-term direction once the seasonal ups and downs are averaged away.
- **Naija analogy:** Lagos traffic getting worse year after year as the city grows, whatever the day of the week.
- **Example:** Ka-band availability trend rising slowly from about 99.24% to 99.40%.

W2D1

👯 Twin files

- **Plain English:** Two data files with the **same name** but different contents, in different folders.
- **Naija analogy:** Two people called "Emeka Okafor" in the same office: check the ID before you transfer money!
- **Example:** `battery_health_daily.csv` in week-03 (clean) vs week-04 (planted faults).

W2D3

🌡️ TWTA and RF drive

- **Plain English:** The TWTA (travelling-wave tube amplifier) boosts the signal before it's sent down. RF drive is how hard it's being pushed.
- **Naija analogy:** A generator: the harder you load it, the hotter it runs.
- **Example:** `twta_thermal_daily.csv` : temperature vs drive, r = 0.59 on levels, 0.40 on changes.

W2D2

🏛️ Unit head

- **Plain English:** The head of your department. They agree your project's scope by email.
- **Naija analogy:** Your HOD signing your project approval form.
- **Example:** Their reply email goes into your final submission.

W2D3

📡 Uplink power

- **Plain English:** How strongly the ground station transmits up to the satellite. It's turned up automatically when rain weakens the signal.
- **Naija analogy:** Shouting louder when the generator next door starts. You shout *because* of the noise.
- **Example:** Uplink power averages about 41 dBm when dry and about 44 dBm in rain.

W1D2

🎚️ Uplink power control

- **Plain English:** The ground station automatically raises its transmit power when rain weakens the signal.
- **Naija analogy:** Talking louder on the phone when the network starts breaking.
- **Example:** `uplink_power_dbm` rises in heavy rain: a response, not a cause.

W2D2

🌍 UTC

- **Plain English:** The world's reference clock. Satellites and telemetry usually record in UTC.
- **Naija analogy:** The "master clock" everyone agrees on, like full-time at Old Trafford: the same moment for the whole world.
- **Example:** The eclipse happens at 21:00 UTC.

W1D1

🕳️ Warm-up NaN

- **Plain English:** The empty values at the start of a rolling result, before the window has enough rows.
- **Naija analogy:** You can't average 7 days of salary when you only started work 3 days ago.
- **Example:** `rolling(7)` gives 6 `NaN` values at the start.

W1D3

🇳🇬 WAT

- **Plain English:** West Africa Time, Nigeria's clock. Always **UTC + 1**, with no daylight saving.
- **Naija analogy:** Lagos time is UTC time plus one hour. ("African time" is another matter 😄)
- **Example:** 21:00 UTC = 22:00 WAT.

W1D1

🎶 Weekly rhythm (day-of-week effect)

- **Plain English:** A pattern that repeats every 7 days because people work Monday to Friday.
- **Naija analogy:** Lagos traffic: wahala on weekdays, free roads on Sunday morning.
- **Example:** Weekdays average about 7 tickets, weekends about 3.4.

W1D3

↔️ Wide format

- **Plain English:** One row per time, one column per group. What charts want.
- **Naija analogy:** The same register as a grid: students across the top, days down the side.
- **Example:** `wide` has 52 rows and 6 zone columns.

W1D2

📏 Window size

- **Plain English:** How many rows go into each calculation. Bigger = smoother but slower.
- **Naija analogy:** How many days of gossip you collect before you believe a rumour.
- **Example:** 3, 7 and 30-day windows on the same ticket data.

W1D3

📏 Y-axis range

- **Plain English:** Where the vertical axis starts and ends. Counts should start at 0; percentages that live near the top may zoom in, if you say so on the chart.
- **Naija analogy:** A "before and after" photo zoomed onto the waist: fair if you say it's zoomed. 😄
- **Example:** Availability on 96–100%; tickets per weekday from 0.

W1D2